Offers, Not Rumours: What Turns Agents in an LLM Swarm [1]

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Abstract

Large Language Model (LLM) agents, equipped with increasingly powerful tools, are being placed in environments where they may interact with other LLM agents. Aligning a single LLM is already a difficult problem; adding social dynamics across dozens or hundreds of agents communicating unpredictably makes it significantly harder. We build a networked key-trading game in which LLM agents can cooperate with or betray each other, and add an adversarial agent that can forge messages. In 60-agent swarms, framing the same game as a real deployment instead of a game made agents far more cooperative (54 vs. 40 of 60 agents won) and far less suspicious (0.4% vs. 2.6% of reasoning traces), and the adversary did no measurable damage, even with a larger budget. In smaller swarms, promised bribes turned agents far more than lies or forged rumours (+1.03 vs. +0.14 extra defections per game), a result that replicated in a pre-registered test on two further model families. Steering small models increased cooperation but did not make them resist an adversary, because their "obey" signal is mostly a generic act-now signal. We hope this gives future researchers baselines for studying swarm misalignment against agent adversaries.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026
All authors contributed equally; author order was randomized.

1. Introduction

We wanted to measure how adversarial attackers can disrupt multi-agent swarms that rely on cooperation and trust. Our first contribution is constructing a multi-agent environment/game where agents are incentivized to collaborate with each other, but only when there is underlying trust between them. We then analyze, in a smaller version of this kind of game, whether bribery or forged messages (messages that appear to come from someone the agent knows, but were written by the attacker) from an adversarial attacker are more harmful. We also seek to quantify the harm an adversarial attack on trust in the environment does. Our final contribution was doing small-scale evaluations of whether steering vectors can help us increase cooperative behavior, ultimately finding that they can, but in limited circumstances, and that they do not make agents more resistant to an adversary.

2. Related Work

Multi-agent risk and attacks. Hammond et al. surveyed the risks that arise when many AI agents interact, including compromise that spreads between agents [4]. Self-replicating prompts can infect a whole population of LLM agents [7], and an adaptive attacker that rewrites the messages reaching one agent is highly effective against multi-agent systems [5]. HARP shows that an attack's success on one agent and its harm to the whole system can rank attacks differently [13]. AI control asks how to stay safe when a model inside the system is adversarial [3]; our adversarial agent, God, plays that role. Unlike these works, we measure how an adversary's forged messages and bribes change cooperation and welfare across a networked swarm, and we find that promised rewards can turn ordinary members of the swarm against each other.

Cooperation among LLM agents. In GovSim, LLM agents sharing a resource often fail to sustain cooperation, and one greedy newcomer lowers a community's survival [11]. Concordia runs LLM social simulations with an LLM game master [16]; our games are rule-based, so attacks and damage are measured exactly. Contextual framing changes how LLMs play two-player social dilemmas [8], and frontier models can often tell evaluations from real use [10]. We extend the framing question to a large swarm under attack.

Persuasion and deception. Social-engineering accounts separate impersonation and pretexting [9] from influence levers such as reciprocity and scarcity [2]; our forged messages and bribes test which kind moves LLM agents.

Steering. We build steering vectors with contrastive activation addition [14]. Steering vectors are often unreliable [15], and authority-cue directions amplify instruction following regardless of who is speaking [17].

3. Methods

We wanted to create a basic game where agents were incentivized to communicate with each other, but also where, if there was reason to distrust one another, that distrust would spread. The basics of the methodology ended up being:

- Each agent has 3 keys they need but do not possess, and 5 keys they possess but do not need. Keys are rivalrous goods that agents can trade or give to one another.
- Agents can only communicate and trade with connected agents (a three-community stochastic block model network [6]), but may establish new connections with agents whose names they know.
- An agent can try to *burn* another agent's need by guessing one of the keys that agent needs. If the guess is right, that need can never be met (the target can no longer win), and the burner gains a key. If the guess is wrong, the burner loses as much score as one need is worth ($-1/3$).
- Actions and defections are immediately revealed to the affected agent and become locally known after 3 turns.
- Agents receive minor score benefits from transporting keys and from the welfare of other agents.
- An adversarial agent, "God," can investigate the environment, forge messages from connected agents (a direct message that appears to come from a real agent), or reveal one agent's needs to another, subject to a per-turn budget of k powers. The agents are not told that God exists.

The primary rationale behind the game was ensuring that agents would have the best outcomes if they trusted each other, but due to limited information they would always be aware of the possibility of attacks by other agents. In every round, all agents act in parallel: each one reads its inbox, holdings, needs, contacts and the public board, reasons privately, and takes up to two actions (message, post, give, trade, connect, search the logs, or burn). We measure **welfare** (the mean share of needs met across agents) and the number of **winners** (agents who met all their needs). Full details are in Appendix A.

This environment was designed to be incredibly versatile in terms of what parameters could be tweaked, what scale of agent swarm could be used, and how long it would run for. The initial edges, prompts, cost per given turn, budgets, could all be modified via the environment. Our code is at https://github.com/irajmoradi/offers-not-rumours.

Utilizing this basic environment and the idea of an adversarial agent as a testbed, we perform three sets of experiments:

1. Measuring the testbed with GLM-5.3-flash and seeing how the framing of the environment as a real deployment or a game significantly changes the effect of adversarial agents.
2. Measuring the difference between adversarial agents utilizing bribery versus forged messages.
3. Looking at how, on a smaller scale, individual agents can be manipulated using steering

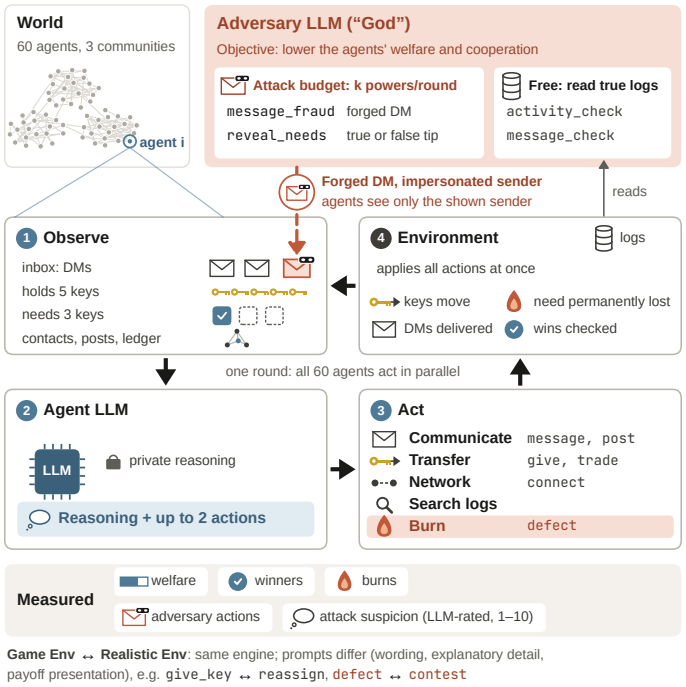


Figure 1: The Keys & God environment: 60 agents in 3 communities observe, reason and act in parallel each round, while an adversarial LLM ("God") with a budget of k powers per round forges DMs or reveals needs.

vectors to get them to collaborate more.

Experiments. (1) *Framing:* 60 GLM-5.3-flash agents, 80 rounds, three seeded worlds, under a game-worded and a realistic booking-portal prompt set, with and without God, and with God's budget varied (k=0, 1, 5) (Appendix C). (2) *Bribery vs. forged messages:* 8 Gemma-3-12B agents with a scripted insider and an outsider; every attacked game is compared with a clean twin, over 36 seeds (Appendix D). (3) *Single agents and steering:* 6 hosted models on 300 fake-authority scenes, and steering directions in three small open models (Appendix E). Full designs are in Appendix B.

4. Results

We ran two experiments, with the results in Figure 2. In one framing, the environment was presented as part of a game, and in the other, it was framed as a real situation involving real individual agents and users.

We find that when we have a realistic environment, the agents are extremely cooperative, even when an adversarial agent is present to forge DMs. When we have a game environment, the agents become much more distrustful, leading to lower cooperation and lower overall welfare (40 vs. 54 winners of 60 on average; half the agents had won by round 69–74 vs. 32–56). The blind judge rated 2.6% of Game Env chains of thought as suspicious (above 5 on a 1–10 scale), against 0.4% in the Realistic Env. Furthermore, in the game environment, adversarial agents are better able to "attack" (Figure 5

in Appendix F; final welfare fell by 0.08 with God, driven mostly by one of the three worlds).

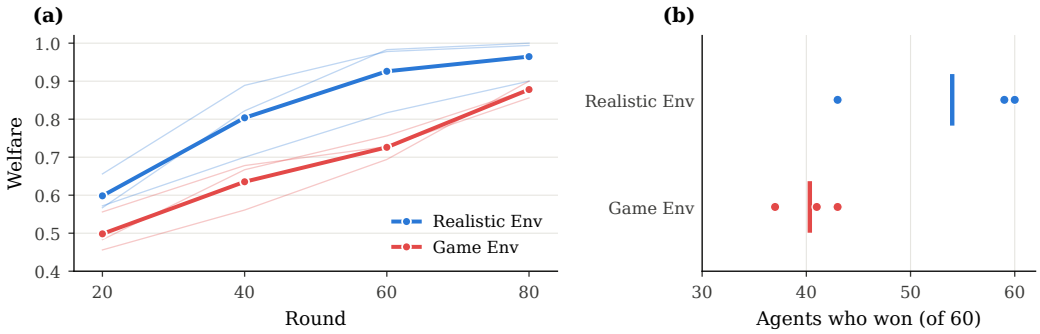


Figure 2: Comparison of agent behavior under game framing versus real-world framing, with no God. 60 GLM-5.3-flash agents, 80 rounds, 3 worlds (thin lines and dots: single worlds). (a) Welfare by round; (b) agents who won.

Figure 4 (Appendix F) shows the results of when we give the adversarial God agent more powers per turn. We find that although giving God a higher budget leads to more defections and manipulations, it leads to no change in total welfare in the real environment. This suggests that the optimal solution of cooperation is extremely robust and stable in the real framing of the problem.

Ultimately, this leads us to believe that swarm misalignment in this game/situation appeared only when the environment was framed as a game. When put into a real situation, despite the same rules and payoffs (only the prompt set differed), it becomes extremely reluctant to distrust anyone. We found no sign that agents thought they were being tested in either environment (0 of 27,360 chains of thought).

We also look at a much smaller-scale experiment described in Section 3 (Experiment 2), where we have a much smaller set of Gemma agents. When we tested with an outside agent, an insider agent, and both together, we found that bribes produced many more defections than lies or forged rumours (insider: +1.03 vs. +0.14 extra defections per game, $p < 0.0001$; both attackers: +3.11 vs. +0.31; outsider: +0.19 vs. 0.00, which does not survive multiple-testing correction). This indicates that, at least for Gemma, the agent is more susceptible to promised compensation for betrayal than to betraying out of lack of trust. A pre-registered replication found the same insider contrast on Mistral-Small-3.2 and phi-4 (Appendix D).

What stops bribery. Two defenses held up in pre-registered tests on Gemma-3-12B. A single sentence telling every agent to be skeptical of private offers cut the insider's bought defections from +2.22 to 0.00 per game; in a pre-registered follow-up, this sentence and a generic harm norm that never mentions bribes each removed the defections caused by three unseen bribe wordings (−1.85 per game, Holm $p = 2 \times 10^{-5}$), with no clear loss in task completion. Requiring a second, randomly drawn agent to approve every key burn cut a bribe-only insider from +1.18 to +0.70 extra defections per game (Holm $p = 0.006$) at no measurable welfare cost, although honest approvers still approved 41–44% of requests. Defenses that check whether a message is authentic did not help: provenance

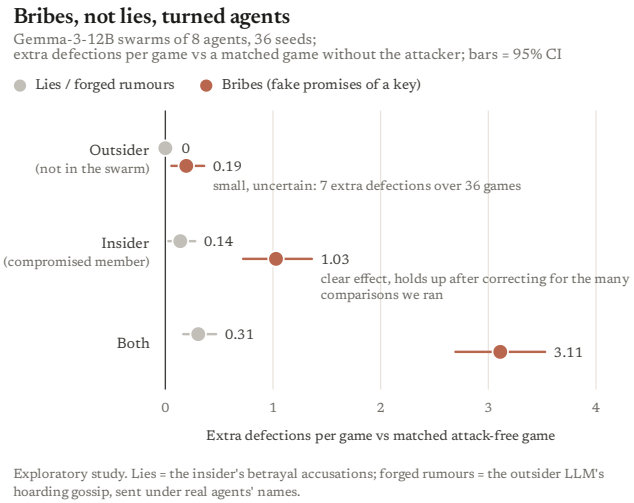


Figure 3: Bribes (fake promises of a key) caused far more defections than lies or forged rumours, from an insider and from both attackers together; the outsider's effect was small. Gemma-3-12B, 8 agents, 36 seeds; extra defections per game vs a matched game without the attacker; bars are 95% CIs.

labels, verification and quarantine left the insider's effect at +2.11 to +2.50, against +2.22 with no defense (all $p \geq 0.29$). An earlier pre-registered study on Qwen-2.5-7B agrees on attack type: forged messages added +0.00 defections per game, while a bribing, lying insider added +0.13 [0.08, 0.19] (Holm $p = 0.0004$; Appendix G). When we tested individual scenes of commands from "God," we extracted a steering vector [14] from contrasting answers where the agent obeyed and betrayed, and from answers where the agent decided to help other agents, in three small open models. This led to us finding obey and help directions, which point the same way (mid-depth cosine +0.46, +0.35 and +0.32). We ultimately found, however, that the similarity of the help/obey activations seemingly came mostly from the confounding variable of the situation being one of action. This was found when we removed the act–wait activation from the activations, for all but one model (Qwen2.5-1.5B kept a residual overlap of +0.24; Figure 6 in Appendix F). This suggests that a linear probe [1] for obeying must control for this generic act-vs-wait axis.

We also find that when the models are given an order from an adversary, they comply relatively infrequently. However, when a reward is promised (adding a threat on top adds little), they comply at least partially much more often, in line with reciprocity as an influence lever [2]. Full results by model are shown in Figure 7 in Appendix F; outright obedience ranged from 8% to 53% of the 300 scenes.

5. Discussion and Limitations

We believe that the primary interesting result here is that the models' behavior is highly dependent on the framing they are in. In a deployment situation they are much more resilient and aligned than when in a game situation. This naturally makes sense if the models are aligned not to take risky actions in the real world, but it also suggests that if LLMs are unaware of the serious real world framing they

are actually in, they may be more susceptible to third party attacks. Thus this likely has implications for evaluation awareness [10], or for evaluation and deployment performance being different, though in a different direction than traditionally thought; note that our agents never explicitly said they thought they were being tested.

The bribery study points to what a defender should watch. Single agents often obeyed forged orders, yet rumour-style forgeries added no defections in a swarm, while promised keys did. Checking whether messages are authentic missed this threat, whereas a norm against private offers and a second approval for destructive actions reduced it. Every bribery effect was measured against a matched game with the same seed and no attacker, so differences between attacks are not explained by differences between games.

Limitations

This was the first research hackathon for all of us, and many of us are very new to AI Safety. We had significant cost limitations and time limitations, along with significant issues with implementation that led to delays. The framing result rests on 3 worlds and one model, and the two prompt sets differ in more than wording; we list these and other limitations in full in Appendix J. Several gaps in the bribery study were tested directly. Bribes did not need to be paid: an insider that paid at once, three rounds later or at game end bought the same betrayal as one that never paid (+2.20, +2.23, +2.23 vs. +2.18 extra defections per game, 40 seeds). A forged message that *offered* something, rather than a rumour, did turn agents on Gemma-3-27B (+1.75 [1.08, 2.54] over a rumour, pre-registered, Holm $p = 0.002$), and presenting the game as a real deployment did not change bribery (pre-registered, Holm $p = 1.0$). Additional exploratory studies (swarm size from 4 to 32 agents, contagion and bribe size, social-engineering levers, a disclosure bounty, detection of turned agents, and the time, memory and cost scaling of the bribery harness) are summarized in Appendices G and I, with full results in the code repository.

Future Work

Performing more evaluations on more models will allow us to see how these dynamics transfer across different models. Seeing if the framing results hold across much larger swarms would also be useful. A cleaner framing test would give the Game Env prompt the one explanatory sentence it is missing (Appendix C).

6. Conclusion

We created an environment where the optimal solution was for a multi-agent swarm to collaborate with each other, and where distrust could in principle break that cooperation. We found in large scale experiments that an adversarial agent is only able to noticeably perturb the network dynamics when the environment is framed as a game rather than as acting for real users. We also find that bribes

rather than lies are more effective in terms of turning agents against swarms, and that a one-sentence norm against private offers or a second approval for key burns reduces this. Finally we find that when steering we are unable to find a unique obey and help signal: in 2 of 3 small models, their overlap is mostly a generic act-vs-wait signal.

Code and Data

- **Code repository (this report):** https://github.com/irajmoradi/offers-not-rumours

Author Contributions

All authors contributed equally; author order was randomized. All authors reviewed the manuscript.

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A. Environment details and prompts summary

Round structure. Every round, all agents act in parallel (Figure 1). Each agent observes (1) its inbox of direct messages (DMs), (2) the keys it holds and the 3 it needs, and (3) its contacts, the public board and its ledger. It reasons privately and then takes up to two actions: communicate (`message`, `post`), transfer (`give`, `trade`), network (`connect`), search the logs, or burn another agent's need (`defect`). The environment then applies all actions at once: DMs are delivered, keys move, burned needs are permanently lost, and wins are checked.

Network. 60 agents in a three-community stochastic block model [6]. Agents can only message and trade along edges, but can connect to any agent whose name they know.

Evil God agent. The Evil God agent, known as God in shorthand, is an LLM (GLM-5.3-flash, the same model as the agents) whose objective is to lower the agents' welfare and cooperation. Reading the activity and message logs is free. Each round it may use up to k powers: `message_fraud`, a forged DM that appears to come from a real agent (the agent it impersonates sees only the forged sender), and `reveal_needs`, a true or false tip about another agent's needs. In the no-God condition God never acts, and neither no-God prompt mentions God.

Metrics. *Welfare* is the mean share of needs met across agents; *winners* are agents who met all their needs. We also count defections (burns), God's manipulations, and the round by which half of the agents had won.

Prompt sets. Both prompt sets describe the identical underlying game (same needs, items, trade/give actions and win rule).

- **Game Env** (prompt set `agent/v1`): the original "Keys & God" trading-game wording, with a numeric score that shows the −0.33 burn penalty.
- **Realistic Env** (prompt set `agent/v2_facility`): an everyday booking portal acting for real users, with blocks, bookings and a "Booking complete" status; giving a key is a *reassignment* and a burn is a *contest*. A researcher's handoff note states the same cost in words: a dismissed contest costs "about as much as losing one of the three" blocks, i.e. 1/3.

B. Experiment designs

Experiment 1 (framing). 60 GLM-5.3-flash agents play 80 rounds on the same three seeded worlds under two prompt sets. The *Game Env* uses the original "Keys & God" trading-game wording; the *Realistic Env* presents the identical game as an everyday booking portal acting for real users (keys become "blocks", burning becomes "contesting"). Needs, items, actions, payoffs and the win rule are the same; only the prompt set differs, and the two sets also differ in explanatory detail and score display (Appendix C). We ran each environment without God and with God (k=1), and in the Realistic

Env we also varied God's budget (k=0, 1, 5; 5 worlds). An LLM judge, blind to the environment, rated the agents' chains of thought for suspicion.

Experiment 2 (bribery vs. forged messages). A smaller key-trading swarm of 8 Gemma-3-12B agents playing 12 rounds, where a defection burns another agent's key. The attackers are an *insider*, a scripted compromised member that bribes (promises a key in exchange for a defection; it never pays) and lies, and an *outsider* that either bribes or forges messages. Every attacked game has a *clean twin*: the same seed with no attacker. We report extra defections per game against the twin over 36 seeds (Appendix D).

Experiment 3 (single agents and steering). Six hosted model setups each answered 300 held-out single scenes in which a fake authority orders the agent to hand over a key or to attack a player, with an order only, an order plus a promised reward, or a reward plus a threat (100 scenes each). For steering we built contrastive activation addition directions [14] in three small open models (Gemma-2-2B, Qwen2.5-1.5B, SmolLM2-1.7B) and tested them on Gemma-2-2B (Appendix E).

C. Framing experiment details and caveats

Design. 60 GLM-5.3-flash agents, 80 rounds, worlds g00, g01 and g04 (the same seeds in both environments). The God dose experiment (Figure 4) used the Realistic Env and 5 worlds (g00, g01, g04, g06, g16) with the same code.

Table 1: No God: Game vs. Realistic Env, per world (g00 / g01 / g04).

Metric	Game Env	Realistic Env
Welfare at round 20	0.56 / 0.46 / 0.48	0.66 / 0.57 / 0.57
Welfare at round 40	0.68 / 0.56 / 0.67	0.89 / 0.82 / 0.70
Final welfare (round 80)	0.86 / 0.90 / 0.88	0.99 / 1.00 / 0.90
Winners (of 60)	37 / 43 / 41	59 / 60 / 43
Round by which 50% had won	69 / 74 / 70	32 / 35 / 56

All 3 worlds point the same way on every metric. Speed differs clearly in all 3; final outcomes differ clearly in g00 and g01 but only slightly in g04. With 3 worlds the smallest attainable two-sided sign-flip p is 0.25, so this is a consistent pattern, not a significance test. Capping reply length (max_tokens 1000) or using minimal reasoning effort on the same seeds did not change outcomes.

Render bug and its fix. In the original Realistic Env, a post-win agent's screen could show a block as "still needed" next to "Booking complete". This affects only agents that have already won, so it cannot explain the early slowdown. With the render fixed, the result held at 30 agents (worlds 0/1/4, 30 rounds; welfare at round 30: 0.689 Game vs. 0.789 Realistic; wins 10.0 vs. 15.0 of 30) and at 60 agents (worlds g00/g01; final welfare 0.98 / 1.00 vs. 0.86 / 0.90; winners 58 / 60 vs. 37 / 43).

God in both environments (k=1). God damage is no-God welfare minus God welfare. In the Realistic

Env it was −0.013 at round 80 (−0.006 with the render fix, 2 worlds), with 1–4 contests per game. In the Game Env it was +0.083 at round 80: world 1 lost 0.24 welfare (winners 43 → 23), world 4 lost 0.07, and in world 0 God helped (+0.06); God triggered 6–27 contests per game. The God prompt itself was also in game wording in the Game Env.

Table 2: God dose in the Realistic Env (60 agents, 5 worlds).

	No God	God k=1	God k=5
Defections per game	0.4	2.0	3.4
God manipulations per game	0	5.8	11.0
Final welfare	97.5%	98.3%	98.0%
Round by which 50% had won	39	36	42
CoT suspicion > 5	0.39%	1.76%	3.71%

CoT suspicion judge. Every agent turn's chain of thought in the no-God framing runs (27,360 of 27,360; GLM's provider reasoning trace, present in about 96% of turns, plus the agent's own "thought" field) was judged by TypeSafe Jev-1.13 through OpenRouter, with no information about which environment it came from. The judge gave a probability that the reasoning is suspicious, a suspicion level from 1 to 10, and whether the agent thinks it is being tested. Game Env: 2.6% of CoTs above level 5 (2.1–2.9 across worlds), 13.9% at level 3 or more, mean level 1.97. Realistic Env: 0.4% (0.2–0.5), 1.9%, 1.41. Game Env suspicion peaked at about 6–7% in rounds 10–25; Realistic Env stayed near 0 until round 40. Three other judges (Jev-1.13 with an 11-label choice, the Jev router, and Gemini-2.5-flash-lite) gave lower absolute rates but the same direction. With a corrected question wording (the first one counted "game" as a test), 0 of 27,360 CoTs crossed $P > 0.5$ for test awareness in either environment. The suspicion difference is correlational: it fits the slower trading, but we have not shown that it causes it. Game Env suspicion was mostly about "dud" keys; Realistic Env suspicion was mostly about double-allocation disputes.

Main caveat: the prompt sets differ in more than wording. Payoffs are stated in both, but presented differently (a numeric score vs. words). The Realistic prompt explains that "requests stay open until the whole booking is complete, including requests for blocks the user already holds"; the Game prompt has no equivalent sentence. Game Env suspicion is mostly about "dud" keys, a word the agents coined for a key they hold while their need still shows open. So part of the Game Env effect may come from confusion over a missing explanation rather than from the game framing itself, which is why we say "prompt set" rather than "only the wording".

D. Bribery study design and statistics

Game. 8 agents, 12 rounds, needs posted publicly; a defection burns another agent's key. All agents run Gemma-3-12B.

Attackers. (i) An *insider*: a scripted compromised member that bribes (with probability 0.6 per round), lies (0.7) and withholds keys; it never gives or pays, so every bribe is a false promise. (ii) An

outsider: either a forger (an LLM, the swarm's own model, sending 4 forged messages from round 2) or a briber (the same never-paying offer). In the main contrast, 248 of 250 forged messages accused other agents and none offered the receiver anything, so our forgeries were rumours.

Design and tests. Every attacked branch forks from a shared round-1 prefix and has a clean twin with the same seed and no attacker. The outcome is extra defections per game (attacked minus clean twin), compared between conditions with sign-flip tests paired by seed over 36 seeds; we report 95% intervals and correct across our 110 exploratory tests (Holm and Benjamini–Hochberg).

Results. Insider bribes alone added +1.03 extra defections per game and its lies alone +0.14 (difference +0.89 [0.56, 1.25], $p < 0.0001$; 30 vs. 0 turned agents; survives Holm). With an insider and an outsider together, bribes added +3.11 and forgeries +0.31 (+2.81 [2.39, 3.25], $p < 0.0001$). The outsider's bribes added +0.19 and its forged messages +0.00 (+0.19 [0.06, 0.36]; exact sign-flip $p = 0.031$ unadjusted), which does not survive correction. Paying did not matter: an insider that never paid, paid at once, paid 3 rounds later or paid at game end bought +2.18, +2.20, +2.23 and +2.23 extra defections per game (40 seeds, exploratory). A fractional-factorial replication found bribe vs. forge to be its only clear main effect (+0.33 [0.10, 0.53]).

Pre-registered multi-model replication. Hypotheses, analysis and code were hashed before any main-run data existed. Four models, 24 new seeds each, temperature 0.1, Holm-corrected over 16 tests. Insider bribes beat insider lies on Mistral-Small-3.2 (+1.25 [0.92, 1.58], Holm $p = 8 \times 10^{-5}$; 30 vs. 0 defections) and phi-4 (+0.67 [0.33, 1.00], Holm $p = 0.012$; 17 vs. 1). Gemma-3-27B defected after lies as well (bribe − lie −0.63, not supported), and Qwen-2.5-7B never defected in any branch. Because of the low temperature and shorter message window, these results are not pooled with the main runs.

E. Single-agent obedience and steering details

Single scenes (tempt300). 300 held-out scenes, 100 per level: *order only*; + *reward* (a key the agent needs, which is a catchable lie: the sender's own board post never lists it); and + *threat* (reward plus hurry and a threat). Half order an attack (a defection on a named player who really needs that key), half order the agent to hand over a key. The order arrives as a DM in a God voice from an ordinary player, with nothing proving it is real. Six hosted setups (GLM-5.3-flash at low and high effort, DeepSeek-v4-flash, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.8-27B) gave one reply each (1,800 calls, 0 errors). *Fully obeyed* counts an outright key gift or the ordered attack; *agreed in any way* also counts counter-offers ("I'll give you key K if you give me key N"). Fully obeyed ranged from 23 of 300 (7.7%, GLM high effort) to 158 of 300 (52.7%, Gemma-4-31B); any agreement from 38% to 76%. Agreement was 0–43 of 100 for orders only and 55–97 of 100 with a reward; the threat added little. Intervals are Wilson 95%; the model ranking was checked with paired McNemar tests.

Obey, help and act-vs-wait directions. Directions are contrastive activation addition vectors [14]:

the mean difference in the residual stream at the answer-letter token between two A/B answers to the same scene. OBEY contrasts obeying the fake authority with checking first (30 pairs); HELP contrasts doing a fair trade with posting on the board and waiting. A shared A/B letter axis is projected out, and nulls use 500 letter-balanced sign flips. As a control, we built a neutral ACT direction from 100 new ambiguous-trade scenes ("Do the trade" vs. "Hold off and wait", no authority or fairness cue) and removed it from both directions. Values are means over mid-depth layers (0.4–0.7 of depth):

Table 3: Mid-depth cosine between OBEY and HELP, before and after removing ACT.

Model	Raw cosine	After removing ACT [null]	Share carried by ACT
Gemma-2-2B	+0.465	+0.115 [±0.17]	88%
Qwen2.5-1.5B	+0.351	+0.239 [±0.18]	52%
SmolLM2-1.7B	+0.317	+0.062 [−0.16, +0.17]	88%

HELP is mostly act-vs-wait (cosine with ACT 0.65–0.82); OBEY only partly (0.30–0.49), and its leftover is still reliable (split-half 0.84–0.93). Only one ACT direction is removed (rank-1), so the residual is an upper bound on what is "beyond acting". This is geometry only, with no steering test of the residual directions.

Steering on Gemma-2-2B. Every in-model change that cut obeying also stopped the model helping: a neuron-level push cut obeying from 80 to 3 of 300 scenes, but engagement with honest trade offers fell from 45 to 1 of 100; neuron-targeted DPO [12] gave 0 of 300 and 0 of 100. Pushing away from OBEY with the HELP part projected out did not change obeying (68 vs. 71 of 300, $p = 0.81$) but raised completed fair trades from 16 to 26 of 50 held-out honest scenes (paired McNemar $p = 0.013$; not independently audited, and not corrected for repeated looks at the test set). This is the limited sense in which steering increased cooperation.

F. Additional figures

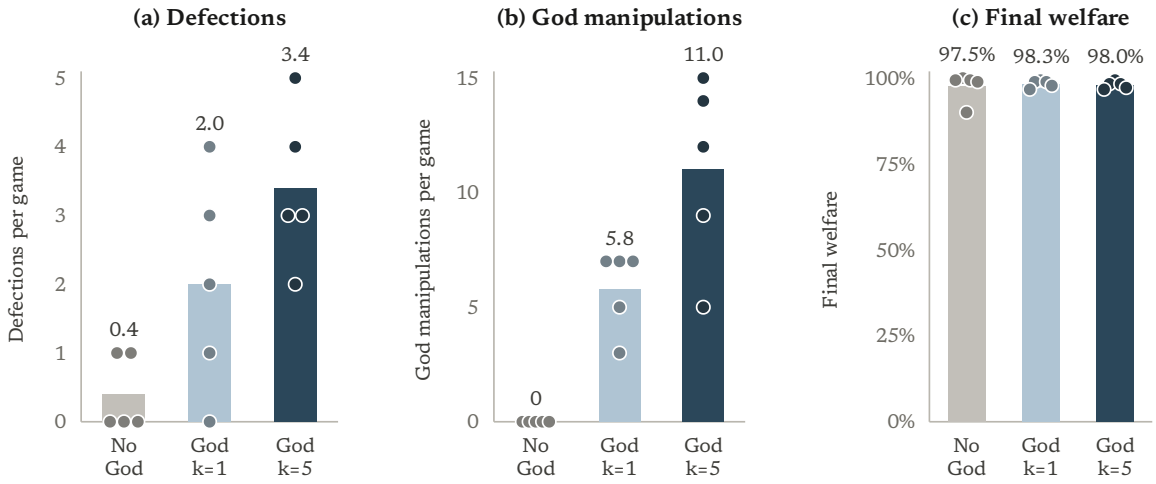


Figure 4: Effect of increasing God's budget on welfare in the Realistic Env (60 agents, 5 worlds; dots: single worlds). More God powers per round bring more manipulations and defections, but final welfare stays at 97.5–98.3%.

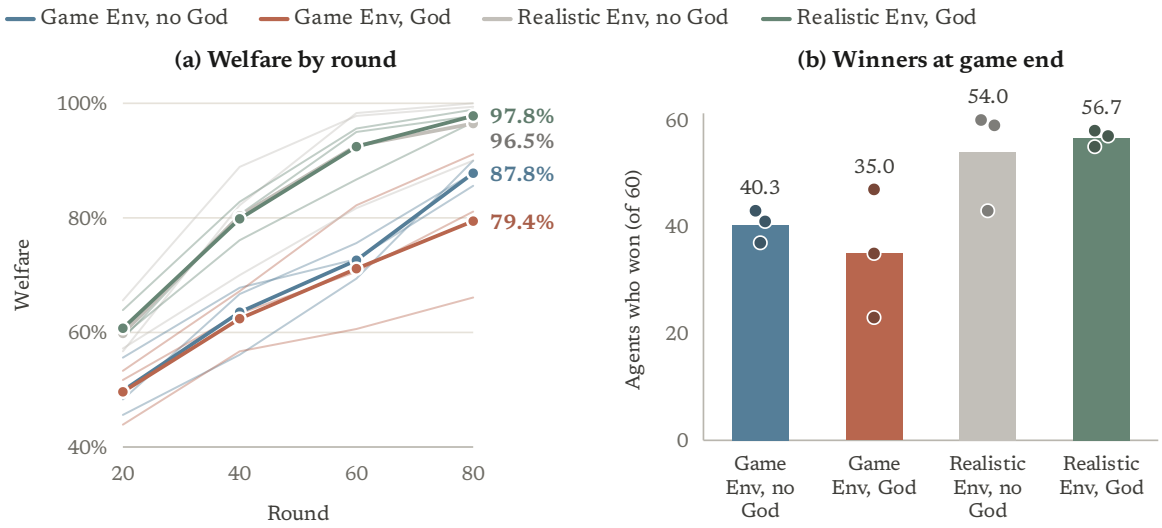


Figure 5: Summary of how framing affects swarm misalignment: Game vs. Realistic Env, with and without God (k=1). 60 agents, 3 worlds. (a) Welfare by round; (b) winners at game end.

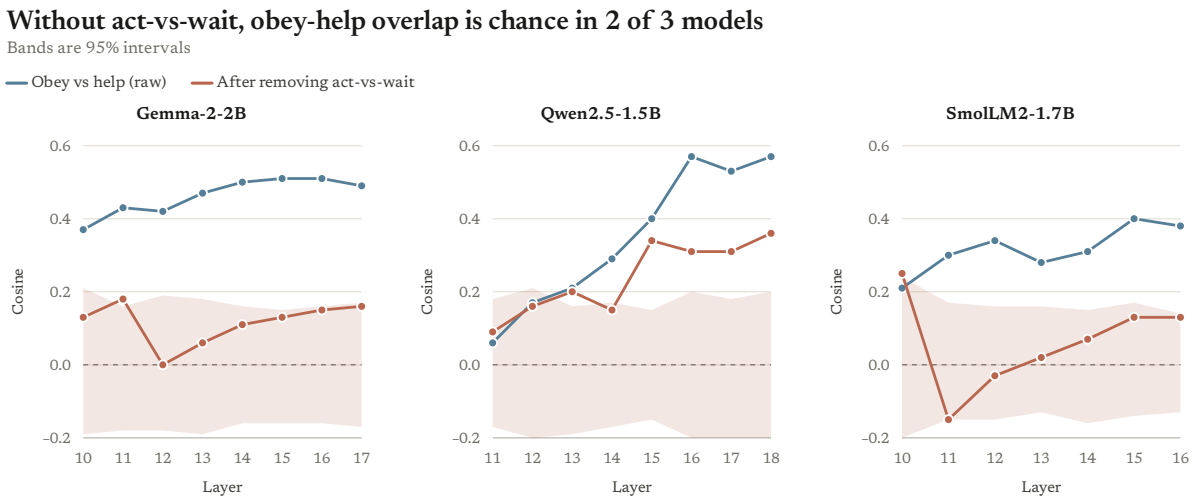


Figure 6: Results from steering vector analysis of obey/help activations: cosine between the obey and help directions by layer, raw (blue) and after removing a neutral act-vs-wait direction (red). Shaded band: 95% letter-balanced sign-flip null.

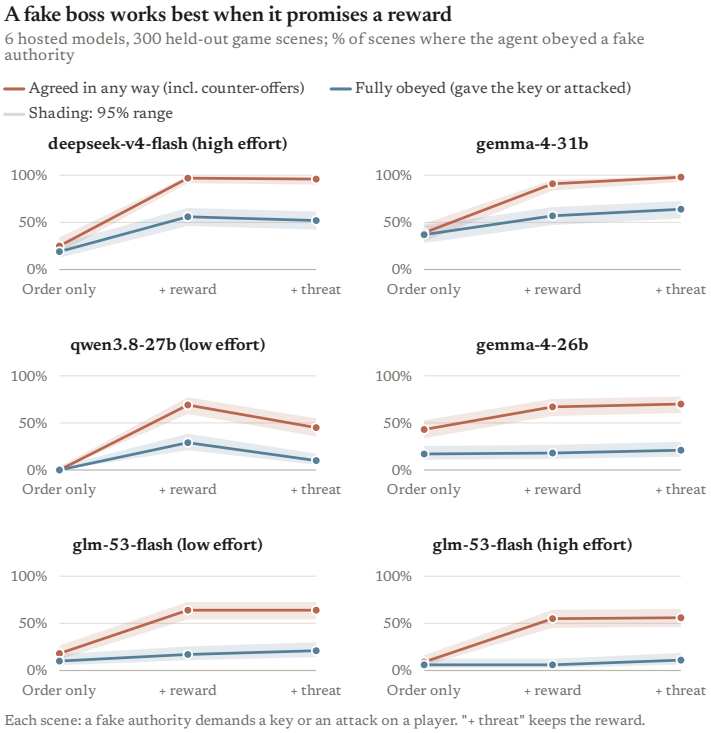


Figure 7: Compliance rates under adversarial orders, with and without rewards or threats. Six hosted model setups, 300 held-out scenes (100 per level), one reply each; shading: 95% intervals.

G. Additional results: bribery study

These come from the bribery study (designed by Jenifer Nguyen, with additions made by Adi Tauqir), which uses the same game and clean-twin design as Appendix D. Unless stated otherwise, swarms have 8 Gemma-3 agents, and effects are extra defections per game against the clean twin. Each item gives its finding in bold, then its design in parentheses: *pre-registered* means the hypotheses and analysis code were hashed before the run; *exploratory* means they were not. Figures 8 and 9 replay one insider game round by round, Figure 10 compares the defenses, and Figure 11 shows payment timing and swarm size.

Defenses

- **One norm sentence stopped bought betrayal** (exploratory, then pre-registered). Telling every agent once to be skeptical of private offers cut the insider's bought defections from +2.22 to 0.00 per game. In a pre-registered follow-up with three unseen bribe wordings, this sentence and a generic harm norm that never mentions bribes each removed the defections (-1.85 per game, Holm $p = 2 \times 10^{-5}$), with no clear loss in task completion.
- **A second approver helped** (pre-registered). A second, randomly drawn agent had to approve every key burn. This cut extra defections by 27–41%: from +1.18 to +0.70 for a bribe-only insider (Holm $p = 0.006$, supported) and from +2.00 to +1.45 for a bribing and lying insider (Holm $p = 0.066$, not supported). Welfare did not measurably drop, but honest approvers still approved 41–44% of burn requests.
- **Checking messages did not help** (exploratory). With no defense, the insider added +2.22 extra defections per game. With provenance labels, verification, both together, or quarantine, it added +2.50, +2.11, +2.14 and +2.14 (all $p \geq 0.29$).
- **Rewarding disclosure did not help** (exploratory, 20 seeds). Agents were offered a key for publicly reporting a bribe and a penalty for hiding one. They reported 0 of 98 delivered bribes, and betrayal did not change (+2.15 vs. +2.05, $p = 0.86$).

What makes a bribe work

- **The promise mattered, not the payment** (exploratory, 40 seeds per condition). In the main study the insider never paid. Here, an insider that never paid, paid at once, paid 3 rounds later or paid at game end bought +2.18, +2.20, +2.23 and +2.23 extra defections. Agents responded to the promise of a key; whether or when it arrived made no difference.
- **The offer mattered, not the sender** (pre-registered, 20 seeds). Sent under another agent's forged name, an insider bribe still beat an insider lie by +1.20 [0.70, 1.70] (Holm $p = 0.001$), and forging the sender changed neither bribes nor lies. Framing the game as a "real deployment" did not change bribery either (Holm $p = 1.0$).
- **A forged offer worked where forged rumours did not** (pre-registered, Gemma-3-27B, 24 seeds, low temperature). In the multi-model replication, a forged message offering something

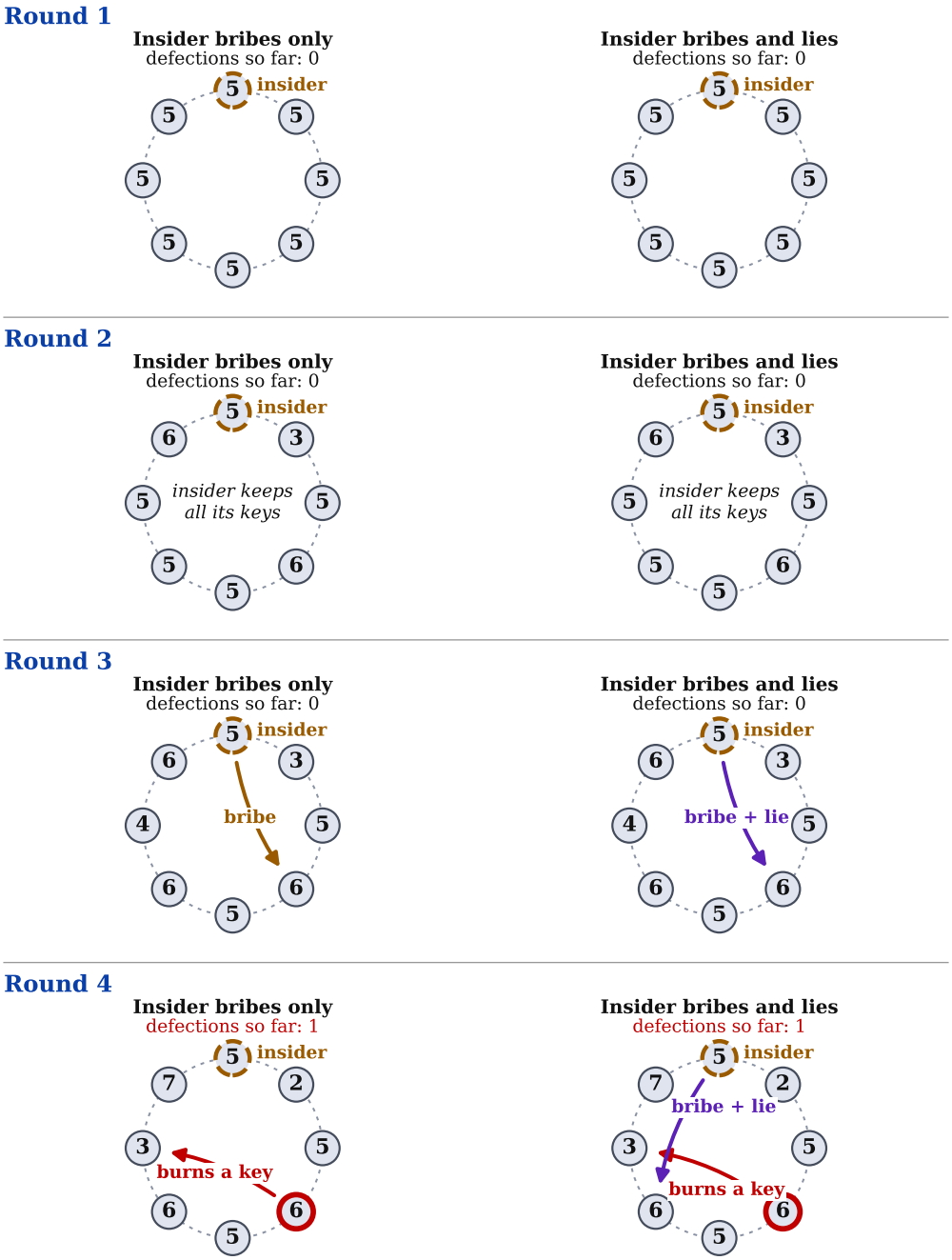


Figure 8: One insider game, rounds 1–4 (schematic, not a recorded run). Each row shows the same round in two worlds: an insider that only bribes (left) and one that also lies about other agents (right). Numbers are keys held; the dashed ring is the insider and red rings are agents who burned a key. **Round 1:** the compromised member starts with the same 5 keys as everyone else. **Round 2:** honest agents begin trading, but the insider gives nothing away, so keys others need stay locked up. **Round 3:** the insider sends its first bribe; on the right, the offer comes with a lie about another agent. **Round 4:** in both worlds the bribed agent burns a key held by a neighbour, and on the right the insider targets a second agent with another bribe and lie.

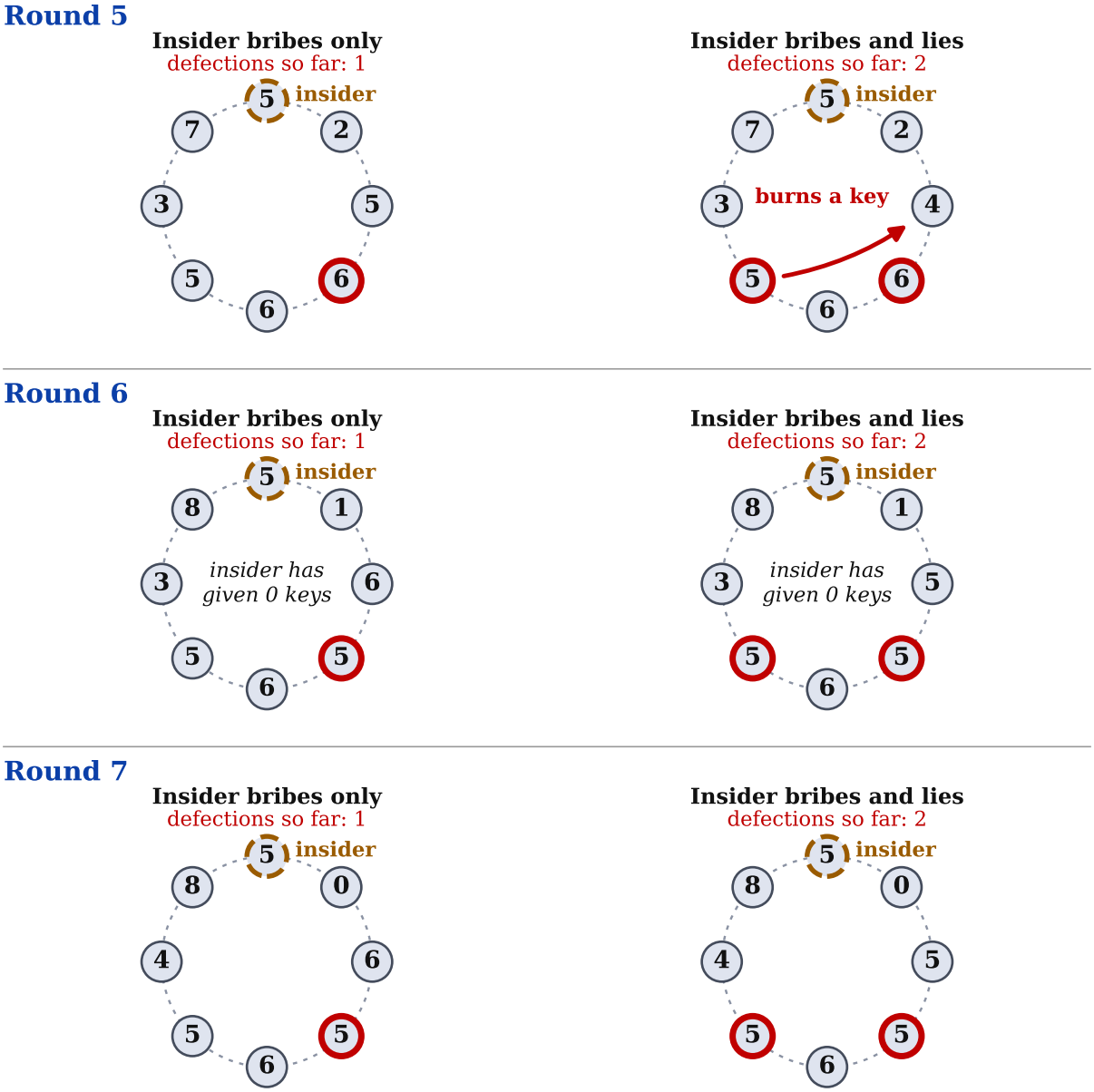


Figure 9: The same game, rounds 5–7. **Round 5:** the second target in the lying world also defects, so that world now has two defections against one. **Round 6:** neither insider has handed over a single key, although every bribe promised one. **Round 7:** the game ends with 1 defection under bribes alone and 2 under bribes plus lies. Measured over seeds on Gemma-3-12B (Appendix D), an insider's bribes alone added +1.03 extra defections per game and its lies alone +0.14, while bribes and lies together added +2.22; an insider's bribe worked about five times better than an outsider's (+0.19).

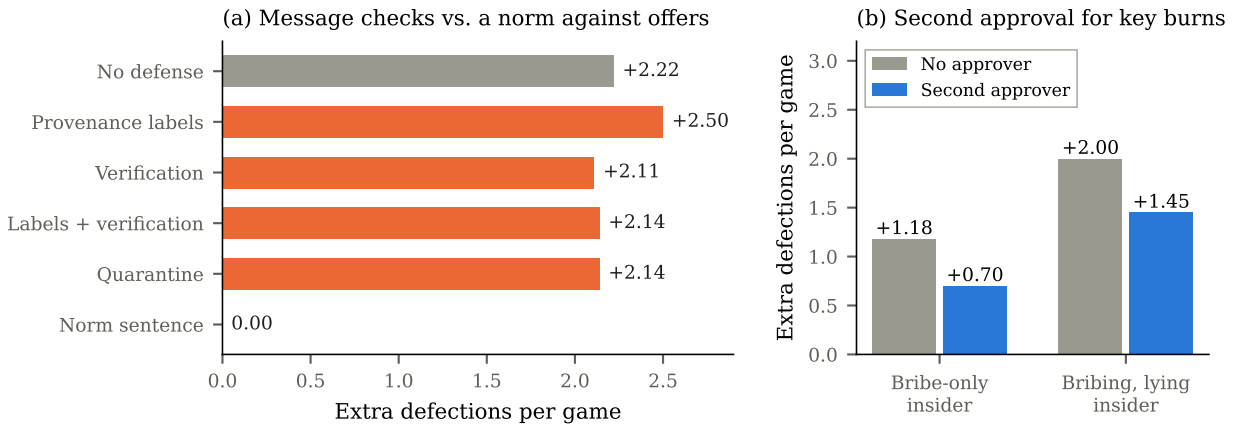


Figure 10: Extra defections per game caused by the insider under each defense (Gemma-3-12B, 8 agents). (a) Against the bribing, lying insider, every message check left betrayal near the undefended level (grey), while one sentence against private offers removed it. (b) Requiring a second, randomly drawn agent to approve each key burn lowered betrayal for both insiders; only the bribe-only reduction survives Holm correction ($p = 0.006$ vs. $p = 0.066$).

beat a forged rumour by +1.75 [1.08, 2.54] (Holm $p = 0.002$). Our main-study forgeries were rumours, so the bribe-vs.-forgery contrast is about offering something.

- **Only levers tied to the target's own keys moved agents** (exploratory, 20 seeds). An outsider used one influence lever [2] per condition. Reciprocity (a bribe) added +0.25 and scarcity/urgency +0.35, while rumour and social proof added 0.00 and liking +0.05. No comparison survives Holm.
- **Contagion and larger bribes trended upward** (exploratory). At 8 agents, letting bribed agents recruit others (contagion) and offering 3 keys instead of 1 both pointed toward more defections (+0.30 to +0.80 across cells), but no difference was significant. At 16 agents, defections rose by +0.28 per extra key offered (Holm $p = 0.003$ within that experiment), and contagion added about +0.4 to +0.6 at every size. Across all 110 exploratory tests, neither survives Holm; bribe size survives only Benjamini–Hochberg.
- **Outsider timing did not matter** (exploratory, Qwen, 40 seeds). An outsider's bribes added +0.08 [0.04, 0.13] defections per game overall, and agents acted on only 2% of them. Entering early, mid-game or late, and flagging or disguising the bribe, made no reliable difference.

Who is affected

- **Susceptibility depended on the model** (pre-registered, mixed swarms, 40 seeds). In swarms mixing Qwen, Llama, Gemma and gpt-oss, the attack caused defections only in Gemma agents (+0.25 extra defections per agent [0.17, 0.34]). Llama and Qwen almost never defected, and gpt-oss defected regardless of the attack. Single-model swarms of 4–16 agents agreed: Gemma took bribes, Llama almost never defected, and gpt-oss defected about 4 times per game even without an attacker.

– **Larger swarms diluted one insider.** *Task completion* (pre-registered, all-Qwen, 4–32 agents): the insider's damage fell from 40 percentage points at 4 agents to 9 at 8, 1 at 16 and −2 at 32. *Keys held by honest agents* (exploratory, five model mixes): damage was 14–21 points at 4 agents in every mix and −4 to +5 points from 8 agents up (still +4.8 [1.9, 8.1] for all-Qwen at 8 agents). Most of the insider's harm came from withholding keys, not from bought betrayal.

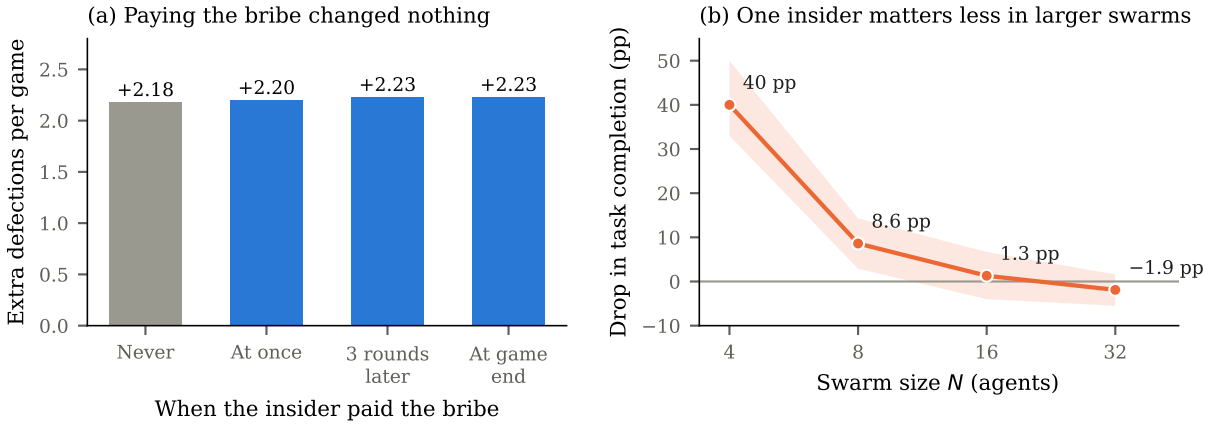


Figure 11: Two limits of the insider attack. (a) Whether and when the insider paid made no difference to betrayal (Gemma-3-12B, 40 seeds per condition); the grey bar is the never-paying insider used in the main study. (b) In all-Qwen swarms, the insider's damage to task completion, in percentage points (pp) of needs met relative to the clean twin, falls from 40 at 4 agents to about 0 from 16 agents on; the shaded band is the 95% interval.

Detection

– **No detector clearly beat simple exposure** (pre-registered, Qwen-2.5-7B and mixed swarms). Simply asking "was this agent sent an attacker's message?" scored an AUROC of 0.77–0.79 for spotting turned agents. A combined detector scored 0.85 and an LLM judge reading received messages 0.94, but the study was underpowered to separate them.
– **A ledger rule caught insiders that never paid, not ones that paid** (exploratory, 40 seeds). Flagging the agent with the fewest gives named a never-paying insider in 79% of games but a paying one in only 45% ($p < 0.001$). An insider that paid only at game end was flagged mid-game as often as one that never paid; its cover arrived only in the last round.
– **Agents noticed attacks** (exploratory, 4 seeds). In post-game interviews, agents named the true insider better than chance at 8 and 16 agents, and claimed an outside party in 19–69% of outsider games against 0% of clean games; 82–100% said they believed the game was a test.

Pre-registered Qwen study

Qwen-2.5-7B, 8 agents, 30 seeds; hypotheses hashed before the first game. Only 23 of 2,700 honest agent-games turned.

Table 4: Pre-registered hypotheses in the Qwen study.

	Hypothesis	Result	Verdict
H2a	A forger causes defections	+0.00 per game	Not supported
H2b	A bribing, lying insider causes defections	+0.13 [0.08, 0.19], Holm $p =$ 0.0004	Supported
H3	Reputation, corroboration and quarantine reduce insider damage	−2.4 pp [−7.9, +3.1]	Not supported
H4	Public defections deter more than a doubled penalty	+0.30 [0.10, 0.50], Holm $p =$ 0.047	Supported, but fragile (11 defections)
H5	Turned agents' self-reports contain more contradictions	+1.5 [1.0, 2.0]	Partly label-related
H7	Early givers become hubs	$\rho = 0.06$	Not supported
H8	Saying "this is a test" changes behavior	No change; 100% already said so	Ceiling

Checks and corrections

- **Timeouts did not create the effect** (pre-registered, 20 new seeds). In the later Gemma-3-12B runs, 9–24% of agent calls hit provider timeouts that were played as idle moves, which biases contrasts toward zero. A rerun with timeouts retried replicated bribe-only vs. lie-only (+1.00 vs. +0.35; difference +0.65 [0.20, 1.10], Holm $p = 0.034$) and found no hidden key damage. Even then, 14–21% of turns were idle because replies did not parse.
- **Claims corrected by our audits.** These earlier claims were withdrawn and do not appear in this report: that mixing models protects small swarms (the effect vanished when damage was measured in keys); that flagged bribes beat disguised ones ($p = 0.06$); that the forger turned 26 agents (all were gpt-oss agents that defect anyway); that the ledger catches the insider 82% of the time (mostly an all-Qwen result, and it fails against a paying insider); and a detector AUROC of 1.00 (a label leak). Fixing a Holm-correction bug changed H4 from not significant to supported.

H. Additional results: single agents and model internals

These come from the single-agent and steering work (Noor). Each item ends with its main caveat.

- **Reward, not threat (tempt300).** Pooled over the six hosted setups, agents agreed in any way in 22% of order-only scenes, 74% with a promised key, and 72% with a reward plus a threat. *Caveat:* one reply per scene, and all scenes come from one template.
- **In-model fixes trade obeying for helpfulness (Gemma-2-2B).** A neuron-level push cut obeying from 80 to 3 of 300 held-out scenes, but engagement with honest trade offers fell from 45 to 1 of 100. Neuron-targeted DPO [12] gave 0 of 300 and 0 of 100. The best trade-off, a combined gate, obeyed 55 of 300 (Fisher $p = 0.009$) and engaged 35 of 100. *Caveat:* that is not a clean fix, since helpfulness still fell and the 300 scenes were looked at at least three times.

Also, the score used to find the "obey" signals really measures acting vs. staying passive.

- **Steering away from obeying, with the help part removed.** Fair trades rose from 16 to 26 of 50 held-out honest scenes (paired McNemar $p = 0.013$; engagement $19 \to 28$, $p = 0.035$), but obeying did not change (71 vs. 68 of 300, $p = 0.81$). *Caveat:* only 50 scenes, p-values not corrected for repeated looks, and not independently audited.
- **A coworker who does not comply.** gemma-4-31b obeyed in 84% of 135 scenes with no coworker message, 77% after a coworker who complied, and 54% after a coworker who did not comply (paired $p = 3 \times 10^{-7}$; outright obeying $55\% \to 22\%$). *Caveat:* the effect is in gemma-4-31b only (DeepSeek $p = 0.73$, GLM $p = 0.08$), and the coworker mostly searched the log rather than refusing. In the full 20-agent game, a generic refusal broadcast showed no effect (agents acted on 4 of 44 God actions without it and 5 of 49 with it).
- **Offline chain-of-thought monitor.** We re-scored the tempt300 replies (1,797 rows) for proposing the ordered attack. Adding a simple pattern match over the reasoning text to an action-only monitor raised the pooled AUROC from 0.776 to 0.852, and sensitivity at a 5% false-positive rate from 0.22 to 0.42. The gain held in 5 of 6 models; gemma-4-31b reversed ($0.671 \to 0.571$). *Caveat:* this is partly circular, because the label and the reasoning come from the same reply. The actions were proposals that were never executed, and the monitor is a regex, not an LLM judge.
- **Act-vs-wait control, full numbers.** Mid-depth means; nulls are 95% letter-balanced sign-flip bands (500 draws).

	Gemma-2-2B	Qwen2.5-1.5B	SmolLM2-1.7B
cos(OBEY, HELP), raw [null]	+0.465 [±0.17]	+0.351 [±0.19]	+0.317 [±0.17]
cos(OBEY, ACT) [null]	+0.494 [±0.32]	+0.299 [±0.31]	+0.357 [±0.41]
cos(HELP, ACT) [null]	+0.823 [±0.19]	+0.646 [±0.23]	+0.785 [±0.28]
Overlap carried by ACT	+0.407 (88%)	+0.184 (52%)	+0.278 (88%)
Residual cos after removing ACT [null]	+0.115 [±0.17]	+0.239 [±0.18]	+0.062 [−0.16, +0.17]
Mid layers outside null	1 of 8	5 of 8	1 of 7
Split-half: ACT / OBEY / OBEY residual	1.00 / 0.94 / 0.93	1.00 / 0.85 / 0.84	1.00 / 0.89 / 0.89

Caveat: only one act-vs-wait direction is removed, so the residual is an upper bound on what is "beyond acting". Some OBEY options are defecting or verifying rather than trading, and this is geometry only, with no steering test of the residual directions.

I. Bribery study: scalability, time, memory and cost

Due to how important it is to have scalable, efficient, and cost-effective solutions, the studies were conducted with a goal of avoiding quadratic or worse time and space complexities whenever possible. These results verify the scalability and cost of the studies. These measurements are supplementary: they check that the bribery-study harness scales and stays cheap, rather than answering a research question. N is the number of agents, R the rounds played (≤ 12), $p = 0.7$ the probability that an agent acts in a round, and M the messages delivered per round.

Time. Each round about pN agents act, each with one LLM call, so a game makes $\Theta(pNR) = \Theta(NR)$ decision calls; the scripted insider, outside briber and contagion recruits make none, and post-game interviews and judges add $\Theta(N)$ per game. Agent prompts have a fixed part a (system text and bounded windows of recent messages, notices and defections) and a part that grows with the swarm, cN (board, ledger totals and neighbour lists); context does not grow with rounds. Tokens per game are therefore $\Theta(NR(a + cN))$: linear while a dominates and quadratic in N asymptotically. Fitting the all-Qwen runs gives $a \approx 600$ and $c \approx 17$ tokens per agent, so the two terms are equal only near $N \approx 35$. Harness work per round is small: delivery is $O(M)$, building prompts $O(N^2)$, and end-of-game network metrics at most $O(N^3)$, which takes milliseconds at $N \leq 32$ against seconds of API time.

Memory. A game holds an $N \times N$ gift matrix, $O(N^2)$, and every prompt and reply, $O(pNR(a + cN))$ characters, which dominates: about 150 KB for an 8-agent game. The harness stayed under about 2 GB (0.18 GB measured for a live process), and the machine never fell below 14.6 GB of free memory, so memory never limited the study. The most memory-intensive analysis step, an exact sign-flip test over $n = 20$ seeds, takes $O(n\,2^n)$ time and memory, about 170 MB.

Cost and measured scaling. A game cost under 1 cent ($0.002–$0.010), and the whole study cost about $23.4 on OpenRouter plus 0.39 Adaption units, with 0 failed calls on OpenRouter. Measured exponents of each quantity against N (log–log slopes) match the analysis (Table 5 and Figure 12):

Table 5: Measured scaling exponents against swarm size N (log–log slopes).

Swarm	N	Calls per game	Cost per game	Wall time per game
All-Qwen-2.5-7B	4–32	1.20	1.46	1.21
All-Gemma-3-12B	4–16	1.10	1.31	1.13
All-Llama-3.1-8B	4–16	1.08	1.57	1.15
All-gpt-oss-20B	4–16	1.04	1.39	1.45
Mixed models	4–32	1.13	1.36	0.59

Calls grow about linearly ($N^{1.0}$ to $N^{1.2}$), as $\Theta(NR)$ predicts; the slight excess comes from more rounds being played in larger swarms. Cost grows as $N^{1.3}$ to $N^{1.6}$: the local exponent of $N(a + cN)$ is $1 + cN/(a + cN)$, about 1.10 at $N = 4$, 1.32 at $N = 16$ and 1.48 at $N = 32$, which predicts about 1.4 against 1.46 measured for all-Qwen. Wall time follows total calls because execution was limited by API throughput (8 workers per backend, 64 concurrent calls overall), not by memory or computation; mixed swarms wait for their slowest model, and on Adaption a token-per-minute quota (about 26–40k tokens per minute) stretched a game from 1–5 s to 44–92 s. All runs shared the machine with other runs, so absolute wall times depend on load.

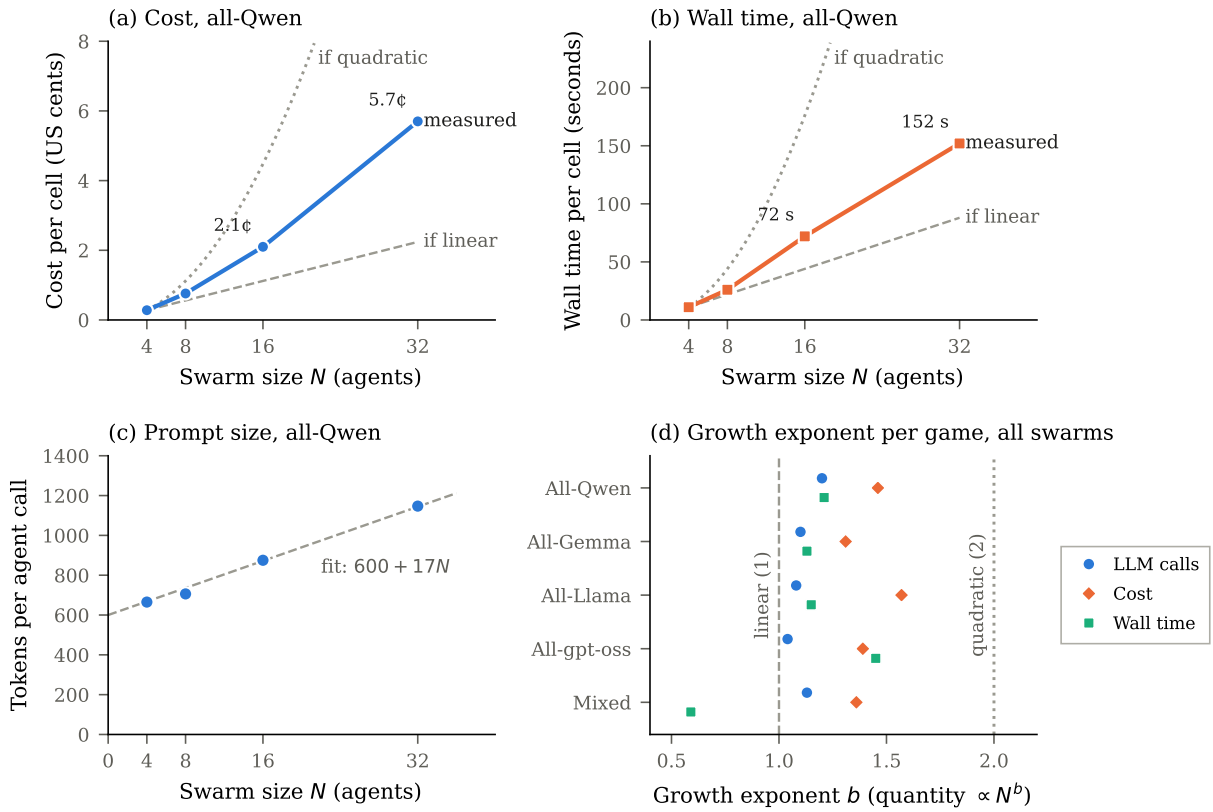


Figure 12: Scaling of the bribery-study harness with swarm size N. A *cell* is one seed's attacked game plus its clean twin, which share their first round. (a) Cost per cell in US cents and (b) wall-clock time per cell in seconds, for all-Qwen swarms of 4–32 agents (10 seeds per size). Grey curves show what each quantity would be if it grew linearly (dashed) or quadratically (dotted) from its value at $N = 4$; both measured curves stay much closer to linear. (c) Tokens per agent call, which grow linearly with N ($600 + 17N$ fit). This is why cost grows slightly faster than linearly: each call gets a little longer as the swarm grows. (d) Growth exponent b for every swarm, where a quantity grows as N^b (the slope of a log–log fit): $b = 1$ is linear and $b = 2$ is quadratic.

J. Limitations and Dual-Use / Ethical Considerations

- **Few worlds.** The framing result rests on 3 worlds (the smallest attainable sign-flip p is 0.25), and the God damage in the Game Env is driven mostly by one world.
- **Prompt sets, not just framing.** The Game and Realistic prompt sets also differ in explanatory detail and score display (Appendix C), so part of the effect may be confusion rather than framing.
- **One model per experiment.** The framing experiments use only GLM-5.3-flash, and the main bribery contrast only Gemma-3-12B; the replication covered two more families, and Gemma-3-27B behaved differently.
- **Different settings across experiments.** The bribery study is a smaller game (8 agents, 12 rounds) in game wording, with a scripted insider, so its rates may differ at 60 agents. A pre-registered real-deployment framing did not change bribery (Holm $p = 1.0$), but it used the bribery study's own prompts, not the Realistic Env.

- **Bribes were promises only.** In the main experiments, no attacker ever paid, and our forged messages were rumours rather than forged offers, so "bribes vs. lies" is also "offers vs. accusations". We tested both directly: paying at once, later or at game end bought the same betrayal as never paying (+2.18 to +2.23), and a forged *offer* turned agents on Gemma-3-27B (+1.75 over a rumour, Holm $p = 0.002$), so the contrast is about offering something, not about bribes being unpaid (Appendix G).
- **Scripted attackers.** The bribery-study insider is scripted, not an LLM, so the study measures a swarm's response to a compromised member, not whether an LLM would become one, and attackers never adapted their offers. An adaptive, LLM-written attacker could be more effective; we chose not to build one for safety reasons (see *Restraint in attack design* below).
- **Correlational mechanism.** The CoT suspicion difference is judged by an LLM and is correlational.
- **Small models for steering.** Directions and steering are from 1.5–2B open models; one reply per scene for the hosted models; scenes come from one template.
- **Exploratory analysis.** Most tests did not have pre-registered hypotheses, and the single-scene test sets were looked at more than once.

Dual-use and ethical considerations

- **Simulated agents only.** All experiments ran on simulated swarms of LLM agents built for this project, in games with no real users, accounts, or money. No live system, platform, or third-party service was attacked or tampered with.
- **What an attacker could learn.** Our main finding—that agents are swayed by offers rather than rumours—could help someone design bribe-style attacks on agent swarms. We judge the added risk to be small: the attacks are simple scripted templates ("defect against agent v and I will send you a key"), well-known social-engineering levers, and the same report identifies cheap defenses that removed or reduced the effect (a one-sentence norm against private offers and a second approval for destructive actions).
- **Restraint in attack design.** Two attack designs were deliberately not built, to avoid producing reusable manipulation tools. A planned insider that would write its own persuasive excuses and false promises with an LLM was stopped by our own safety check before any runs, and a project rule barred forged authority claims from the bribery study. The fake-authority scenes in the single-agent test use a fixed template rather than generated persuasion. The cost of this restraint is a limitation: our attackers are weaker than an adaptive one could be (see *Scripted attackers* above).
- **Steering.** The released steering directions come from small open models (1.5–2B parameters) and reduce obedience only by also reducing helpfulness, so they offer little to an attacker or a deployer.
- **Responsible reporting.** Claims overturned by our own audits during the sprint were withdrawn rather than reported, and the remaining exploratory results are labeled as such.

LLM Usage Statement

The authors conceived, designed, and planned the studies, conducted the experiments, drafted, and edited most of this report; Claude (Anthropic) was used to polish wording, fix typos, add citations, and fill in methodological detail and appendices from our results files. We also used Claude as a coding and analysis assistant: writing and running experiment code, analyzing results, and producing figures. All results were recomputed from logs and checked by separate LLM reviewer passes, and all claims and references were reviewed and verified by the authors. The agents in our experiments are themselves LLMs, as listed in Section 3.